Ideation Phase
Brainstorm & Idea Prioritization Template

Date	09 March 2026
Team ID	
Project Name	Malaria Detection Using Deep Learning
Maximum Marks	4 Marks

Brainstorm & Idea Prioritization Template:

Brainstorming provides a free and open environment that encourages everyone within a team to participate in the creative thinking process that leads to problem solving. Prioritizing volume over value, out-of-the-box ideas are welcome and built upon, and all participants are encouraged to collaborate, helping each other develop a rich amount of creative solutions.

Use this template in your own brainstorming sessions so your team can unleash their imagination and start shaping concepts even if you're not sitting in the same room.

Reference: https://www.mural.co/templates/brainstorm-and-idea-prioritization

Step-1: Team Gathering, Collaboration and Select the Problem Statement

Malaria diagnosis traditionally requires manual examination of microscopic blood smear images by trained medical professionals. This process is time-consuming, requires expertise, and may lead to diagnostic errors. Therefore, there is a need for an automated system that can assist healthcare professionals in detecting malaria parasites accurately and efficiently using artificial intelligence and deep learning techniques.

Step-2: Brainstorm, Idea Listing and Grouping

During brainstorming, the team generated several ideas to solve the problem:

1. Automated malaria detection using deep learning
2. Mobile application for malaria diagnosis
3. Web-based system for uploading blood cell images
4. Real-time parasite detection using CNN models
5. Integration with healthcare databases
6. Using transfer learning models like MobileNet and EfficientNet
7. AI system for identifying infected and uninfected cells
8. Visual interface showing prediction results and confidence scores

Group the ideas into categories.

AI & Machine Learning Ideas

- CNN-based malaria detection
- Transfer learning models (MobileNetV2, EfficientNet)
- Image preprocessing and classification

Application Development Ideas

- Web application for image upload
- User interface for displaying results

- Backend integration with trained model

Healthcare Support Ideas

- Automated malaria screening tool
- Assistance for medical professionals
- Faster diagnosis using AI

Step-3: Idea Prioritization

The team prioritized the idea based on importance and feasibility.

Final Selected Idea:

Malaria Detection using Deep Learning

Reason for Selection

- High impact in healthcare diagnostics
- Feasible using available datasets
- Can be implemented using deep learning frameworks
- Helps automate malaria detection process
- Provides faster and more accurate analysis